Supplementary material for: Temporal quantification of atrial fibrillation burden with distribution-free uncertainty intervals: development and multinational external validation

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Supplementary material

S1. Pre-registration and design decisions

The full analysis plan (candidate topics considered, stress test, novelty map, hypotheses, research questions, splits, seeds, fixed hyper-parameters, endpoints and statistical tests) was committed to version control before any model was trained and is provided as the file *S1_research_design_preregistration.md*. One amendment (A1) was recorded after a one-seed pipeline check and before the full analysis: the emission model of PA-HMMQ would be selected on development patients only (calibration-B and development test patients, LightGBM, mean per-patient absolute error). The rule selected the probability-emission variant (development error 2.858 vs 3.186 pp). The one-seed check also displayed errors for 40-patient subsets of two external cohorts (disclosed in S1). Deviations from the plan: (i) reduced-training-data and window-length analyses omitted Icentia11k for computational reasons; (ii) post-hoc analyses: fair-comparator tests (same emissions, fixed dynamics), per-detector emission selection (R1; selection on calibration-A patients, on which the source dynamics are also fitted, so the selection may be optimistic), exclusion of flutter patients, newly missed patients and tests at 0.1-pp resolution; (iii) one SHDB-AF recording was excluded after the leakage audit (below).

S2. Data processing and leakage audit

Only beat and rhythm annotation files were downloaded and verified against PhysioNet SHA-256 checksums. CPSC2021 records of patient 51 (11 records) were not retrievable from the repository (HTTP 404) and are absent. In Icentia11k, beats outside labelled rhythm regions were treated as unlabelled and excluded. The automated audit hashed every 60-beat window (RR rounded to 8 ms) and fingerprinted every recording. SHDB-AF Data_ID 020 was an exact RR-level duplicate of Data_ID 005 (same Subject_ID; metadata Holter dates 2021 and 2024) and was removed; after removal no recording-level duplicates remained, and 2 individual windows (of 928477) were shared between different patients, consistent with chance identity of very regular rhythms. Patient-level splits were verified disjoint for every seed, and no development patient occurred in an external cohort. The SHDB-AF metadata

columns are Data_ID/Subject_ID (not the names given in the database README); patients with two recordings were aggregated. MITDB paced records were retained as non-AF recordings. Windows separated by an invalid RR interval within a recording were treated as adjacent in the hidden Markov model (valid intervals constituted ≥99.8% of intervals in every database).

S3. Detector details

Features (16). Mean RR; SDNN; coefficient of variation; RMSSD/mean RR; pNN50; pNN20; median |ΔRR|/mean RR; interquartile range/median RR; turning-point ratio; normalised Shannon entropy of the 16-bin RR histogram; Poincaré SD1/mean RR, SD2/mean RR and SD1/SD2; a coefficient-of-sample-entropy-type irregularity measure, computed as SampEn(m = 2, r = 0.2·SD) + ln(0.06) − ln(mean RR) (a hybrid of sample entropy and the COSEn normalisation, not the original fixed-tolerance COSEn); fraction of sign changes of ΔRR; min/max RR ratio. *Classical-LR* used normalised RMSSD, Shannon entropy and turning-point ratio with standardisation fitted on training patients. *LightGBM*: up to 400 trees, learning rate 0.05, 31 leaves, row and column subsampling 0.8, early stopping (50 rounds) on inner-validation patients. *RRNet*: input two channels (log RR minus window mean; ΔRR), three 1-D convolutional blocks (32 channels, kernel 5, batch normalisation, ReLU, residual connections), bidirectional GRU (32 units per direction), mean and max pooling, two-layer head with dropout 0.2; AdamW (learning rate 10^{-3}, weight decay 10^{-4}), cosine schedule, batch 512, ≤12 epochs, early stopping (patience 3) on inner-validation binary cross-entropy; 27,681 parameters. RRNet is a re-implementation inspired by ArNet/ArNet2, not the original code or weights. PyTorch was run with deterministic algorithms; LightGBM used four threads without its deterministic flag, so re-runs may differ negligibly.

S4. PA-HMMQ and interval details

Hidden states $z_t \in \{0 \text{ (non-AF)}, 1 \text{ (AF)}\}$ for consecutive windows of a recording; emissions $e_k(x_t)$ are fixed: in the final variant $e_1 = p_t$, $e_0 = 1 - p_t$ (p clipped to $[10^{-6}, 1 - 10^{-6}]$); in the pre-registered variant e_k is a Laplace-smoothed 50-bin histogram density of logit(p) among calibration-A windows of class k. The source transition matrix and initial distribution are add-

one-smoothed frequencies of reference window labels in calibration-A sequences. For each patient i, expectation–maximisation alternates forward–backward smoothing (all recordings of the patient share A_i, π_i) with the MAP updates $A_i(k,l) \propto \Sigma\xi(k,l) + \kappa A_{src}(k,l)$ and $\pi_i(k) \propto \Sigma\gamma_1(k) + \kappa\pi_{src}(k)$, until the relative log-likelihood change is $<10^{-6}$ ($\leq$50 iterations). Burden is $\Sigma d_t\gamma_t(1)/\Sigma d_t$. Convergence is assessed on the summed log-likelihood over all patients of a cohort. EMQ is applied to isotonic-recalibrated posteriors (calibration A) with the duration-weighted source prior. Posterior samples of the whole state path are drawn by forward-filtering backward-sampling at the converged parameters; $\hat{\sigma_i}$ is the standard deviation of the duration-weighted sampled burden. EMQ is recovered by removing temporal dependence and HMM-src by fixing $A_i = A_{src}$. Complexity: one forward–backward pass is $O(T \cdot S^2)$ with S = 2 states and T windows (≈1,500–3,000 per 24 h), repeated for each EM iteration; FFBS adds $O(K \cdot T)$ for K = 200 samples. The implementation vectorises across sequences by padding to the longest sequence, so memory is $O(N_{seq} \cdot T_{max} \cdot K)$ during sampling; the measured peak Python memory for a 24-h recording was 1.4 MB. The Monte-Carlo error of $\hat{\sigma}$ (200 samples) was not varied; $\hat{\sigma}$ only scales the nonconformity score, whose quantile is calibrated empirically. Timings are CPU wall-clock medians on 4 threads; energy was not measured.

S5. Supplementary tables and figures

Supplementary Tables S1–S14 and Figures S1–S8 follow. All values are generated automatically from the result files in the code repository.

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Table S1 Databases: sources, versions, licences and window counts

Database	Role	Country	Source/ version	Licence	Recordings	Patients	Hours analysed	Windows (60 beats)	Patients with AF	Patients with AFL	AF windows (%)
Long-Term AF Database	development	USA	PhysioNet ltafdb 1.0.0	ODC-By 1.0	84	84	1897	149760	83	0	58.1
MIT-BIH AF Database	development	USA	PhysioNet afdb 1.0.0	ODC-By 1.0	25	25	248	20312	25	8	42.5
MIT-BIH Arrhythmia Database	development	USA	PhysioNet mitdb 1.0.0	ODC-By 1.0	48	47	23	1787	7	3	10.6
MIT-BIH Normal Sinus Rhythm Database	development	USA	PhysioNet nsrdb 1.0.0	ODC-By 1.0	18	18	378	28454	0	0	0.0
Normal Sinus Rhythm RR Interval Database	development	USA	PhysioNet nsr2db 1.0.0	ODC-By 1.0	54	54	1200	92190	0	0	0.0
BIDMC Congestive Heart Failure Database	development	USA	PhysioNet chfdb 1.0.0	ODC-By 1.0	15	15	298	27000	4	0	5.9
SHDB-AF Japanese Holter AF database	external validation	Japan	PhysioNet shdb-af 1.0.1	ODC-By 1.0	97	93	2297	170001	72	12	23.8
4th China Physiological Signal Challenge 2021	external validation	China	PhysioNet cpsc2021 1.0.0	CC BY 4.0	1405	105	471	34706	53	2	35.9
Icentia11k continuous ECG (random subset)	external validation	Canada	PhysioNet icentia11k-continuous-ecg 1.0	CC BY-NC-SA 4.0	790	790	5808	404267	50	25	5.9

All databases from PhysioNet; counts after exclusions (see Supplementary S2). ODC-By, Open Data Commons Attribution License; CC BY, Creative Commons Attribution; CC BY-NC-SA, Creative Commons Attribution-NonCommercial-ShareAlike.

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Table S2 Window-level detector performance (mean ± SD over seeds)

Detector	Cohort	auroc	auprc	sensitivity	specificity	f1	ece	brier
Classical-LR	Development test	0.959 ± 0.012	0.875 ± 0.036	0.851 ± 0.039	0.951 ± 0.009	0.873 ± 0.022	0.058 ± 0.027	0.067 ± 0.008
Classical-LR	SHDB-AF	0.884 ± 0.001	0.628 ± 0.010	0.837 ± 0.023	0.889 ± 0.009	0.763 ± 0.005	0.068 ± 0.010	0.104 ± 0.003
Classical-LR	CPSC2021	0.884 ± 0.004	0.663 ± 0.010	0.926 ± 0.025	0.798 ± 0.018	0.810 ± 0.005	0.100 ± 0.019	0.129 ± 0.005
Classical-LR	Icentia11k	0.961 ± 0.000	0.457 ± 0.011	0.931 ± 0.028	0.933 ± 0.007	0.622 ± 0.009	0.099 ± 0.009	0.057 ± 0.005
LightGBM	Development test	0.992 ± 0.005	0.986 ± 0.009	0.909 ± 0.046	0.983 ± 0.012	0.936 ± 0.023	0.032 ± 0.013	0.035 ± 0.011
LightGBM	SHDB-AF	0.969 ± 0.003	0.902 ± 0.007	0.881 ± 0.011	0.960 ± 0.004	0.878 ± 0.003	0.035 ± 0.012	0.051 ± 0.001
LightGBM	CPSC2021	0.992 ± 0.001	0.980 ± 0.004	0.976 ± 0.009	0.954 ± 0.013	0.949 ± 0.010	0.025 ± 0.007	0.029 ± 0.005
LightGBM	Icentia11k	0.990 ± 0.001	0.776 ± 0.021	0.974 ± 0.006	0.972 ± 0.004	0.804 ± 0.016	0.030 ± 0.002	0.024 ± 0.003
RRNet	Development test	0.992 ± 0.007	0.987 ± 0.012	0.917 ± 0.060	0.983 ± 0.011	0.940 ± 0.029	0.032 ± 0.017	0.033 ± 0.014
RRNet	SHDB-AF	0.973 ± 0.001	0.910 ± 0.005	0.908 ± 0.012	0.952 ± 0.008	0.881 ± 0.006	0.045 ± 0.003	0.051 ± 0.002
RRNet	CPSC2021	0.990 ± 0.002	0.967 ± 0.008	0.989 ± 0.012	0.939 ± 0.023	0.943 ± 0.015	0.036 ± 0.017	0.035 ± 0.009
RRNet	Icentia11k	0.986 ± 0.001	0.686 ± 0.029	0.981 ± 0.012	0.963 ± 0.009	0.762 ± 0.037	0.036 ± 0.010	0.032 ± 0.006

Threshold 0.5. ECE, expected calibration error (15 bins).

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Table S3 Per-patient AF-burden MAE (pp) by estimator, Classical-LR detector

Estimator	Development test	SHDB-AF	CPSC2021	Icentia11k	Pooled external
CC	7.71 ± 0.48	9.94 ± 0.54	17.34 ± 0.81	7.20 ± 0.57	8.54 ± 0.56
CC-MF	7.28 ± 0.52	9.69 ± 0.50	16.95 ± 0.98	6.79 ± 0.57	8.14 ± 0.58
PCC	13.40 ± 1.13	12.31 ± 0.54	22.10 ± 0.95	12.99 ± 0.68	13.90 ± 0.65
PCC-iso	10.87 ± 1.66	11.13 ± 0.35	20.82 ± 0.75	10.55 ± 0.85	11.70 ± 0.74
ACC	7.28 ± 0.91	11.01 ± 0.51	17.75 ± 1.20	6.84 ± 0.34	8.40 ± 0.41
PACC	8.95 ± 1.66	12.16 ± 0.69	20.05 ± 1.37	8.87 ± 0.61	10.37 ± 0.65
EMQ	7.36 ± 2.15	11.42 ± 0.48	18.32 ± 1.30	8.66 ± 0.79	9.95 ± 0.79
HMM-src	7.45 ± 2.16	10.38 ± 0.70	16.94 ± 1.21	9.33 ± 0.91	10.23 ± 0.91
HMM-src[rawE]	6.79 ± 0.83	9.20 ± 0.40	15.76 ± 1.58	6.06 ± 0.66	7.39 ± 0.71
HMM-src[sel] (post hoc R1)	7.54 ± 2.10	10.13 ± 1.02	16.77 ± 1.30	8.63 ± 1.85	9.64 ± 1.68
PA-HMMQ (pre-registered)	7.40 ± 1.78	10.59 ± 0.64	17.50 ± 1.06	9.53 ± 0.94	10.47 ± 0.91
PA-HMMQ (final)	6.86 ± 0.62	9.26 ± 0.39	15.87 ± 1.41	6.10 ± 0.65	7.44 ± 0.69
PA-HMMQ[sel] (post hoc R1)	7.43 ± 1.76	10.29 ± 1.07	16.94 ± 1.55	8.79 ± 1.94	9.79 ± 1.81

Mean ± SD over seeds.

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Table S4 Per-patient AF-burden MAE (pp) by estimator, RRNet detector

Estimator	Development test	SHDB-AF	CPSC2021	Icentia11k	Pooled external
CC	3.55 ± 1.29	4.78 ± 0.46	5.01 ± 1.15	3.72 ± 0.88	3.96 ± 0.83
CC-MF	3.34 ± 1.20	4.52 ± 0.40	4.73 ± 1.21	3.50 ± 0.84	3.72 ± 0.80
PCC	4.02 ± 1.16	5.06 ± 0.50	5.86 ± 1.35	4.10 ± 0.99	4.37 ± 0.93
PCC-iso	3.67 ± 0.82	5.00 ± 0.33	5.50 ± 0.99	3.96 ± 0.67	4.22 ± 0.63
ACC	3.54 ± 1.21	4.97 ± 0.44	4.81 ± 1.26	3.59 ± 0.84	3.85 ± 0.81
PACC	3.71 ± 1.09	5.18 ± 0.51	5.27 ± 1.39	3.82 ± 0.92	4.10 ± 0.89
EMQ	3.03 ± 0.97	4.85 ± 0.44	4.51 ± 1.15	3.37 ± 0.64	3.63 ± 0.61
HMM-src	2.64 ± 1.06	4.55 ± 0.41	4.49 ± 1.01	3.53 ± 0.68	3.73 ± 0.64
HMM-src[rawE]	3.00 ± 1.12	4.29 ± 0.26	4.21 ± 1.14	3.22 ± 0.75	3.43 ± 0.70
HMM-src[sel] (post hoc R1)	3.06 ± 1.04	4.37 ± 0.26	4.68 ± 1.20	3.23 ± 0.82	3.49 ± 0.67
PA-HMMQ (pre-registered)	2.68 ± 1.06	4.60 ± 0.44	4.59 ± 0.95	3.57 ± 0.70	3.78 ± 0.65
PA-HMMQ (final)	3.18 ± 1.08	4.31 ± 0.28	4.20 ± 1.06	3.25 ± 0.76	3.45 ± 0.70
PA-HMMQ[sel] (post hoc R1)	3.06 ± 0.94	4.43 ± 0.33	4.39 ± 1.14	3.27 ± 0.86	3.50 ± 0.78

Mean ± SD over seeds.

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Table S5 Ablation of PA-HMMQ components (LightGBM, MAE in pp)

Variant	Calibration B	Development test	SHDB-AF	CPSC2021	Icentia11k	Pooled external
PA-HMMQ	3.40 ± 1.79	2.97 ± 1.03	4.54 ± 0.36	3.50 ± 1.06	2.84 ± 0.34	3.07 ± 0.30
PA-HMMQ[k=0]	3.35 ± 1.84	2.97 ± 1.03	4.54 ± 0.36	3.54 ± 1.09	2.84 ± 0.34	3.08 ± 0.31
PA-HMMQ[k=100]	3.49 ± 1.76	2.98 ± 1.03	4.54 ± 0.36	3.47 ± 1.03	2.83 ± 0.33	3.06 ± 0.29
PA-HMMQ[k=1]	3.37 ± 1.81	2.97 ± 1.03	4.54 ± 0.36	3.54 ± 1.08	2.84 ± 0.34	3.08 ± 0.31
PA-HMMQ[k=30]	3.46 ± 1.75	2.97 ± 1.03	4.54 ± 0.36	3.48 ± 1.05	2.84 ± 0.34	3.07 ± 0.30
PA-HMMQ[k=3]	3.38 ± 1.80	2.97 ± 1.03	4.54 ± 0.36	3.52 ± 1.07	2.84 ± 0.34	3.08 ± 0.31
PA-HMMQ[rawE,k=0]	2.68 ± 1.56	3.04 ± 1.26	4.23 ± 0.11	2.77 ± 0.38	2.27 ± 0.33	2.51 ± 0.25
PA-HMMQ[rawE,k=100]	2.76 ± 1.53	3.03 ± 1.28	4.24 ± 0.11	2.98 ± 0.45	2.26 ± 0.33	2.52 ± 0.23
PA-HMMQ[rawE,k=1]	2.66 ± 1.55	3.04 ± 1.25	4.23 ± 0.11	2.80 ± 0.38	2.27 ± 0.33	2.51 ± 0.25
PA-HMMQ[rawE,k=30]	2.71 ± 1.54	3.03 ± 1.26	4.24 ± 0.11	2.94 ± 0.42	2.26 ± 0.33	2.52 ± 0.24
PA-HMMQ[rawE,k=3]	2.66 ± 1.57	3.04 ± 1.25	4.23 ± 0.11	2.84 ± 0.38	2.27 ± 0.33	2.51 ± 0.25
PA-HMMQ[rawE]	2.67 ± 1.57	3.04 ± 1.26	4.23 ± 0.11	2.88 ± 0.39	2.27 ± 0.33	2.52 ± 0.24
PA-HMMQ[sel]	2.93 ± 1.82	3.16 ± 1.28	4.29 ± 0.12	3.05 ± 0.47	2.29 ± 0.35	2.56 ± 0.27
CC	2.86 ± 1.25	3.55 ± 1.05	4.51 ± 0.12	3.85 ± 0.76	2.69 ± 0.35	2.99 ± 0.26
EMQ	3.48 ± 1.74	3.10 ± 0.93	4.92 ± 0.52	3.78 ± 1.12	2.72 ± 0.29	3.04 ± 0.21
HMM-src	3.56 ± 1.84	3.00 ± 1.02	4.53 ± 0.37	3.48 ± 1.02	2.82 ± 0.33	3.05 ± 0.29
HMM-src[rawE]	2.84 ± 1.54	3.04 ± 1.32	4.24 ± 0.10	3.02 ± 0.43	2.25 ± 0.31	2.52 ± 0.22
HMM-src[sel]	2.75 ± 1.64	3.13 ± 1.38	4.27 ± 0.14	3.42 ± 0.66	2.26 ± 0.32	2.57 ± 0.21

k, prior strength κ in pseudo-transitions (κ=10 pre-specified); rawE, detector probability used as emission likelihood (final variant); HMM-src, no per-patient adaptation ($\kappa \to \infty$); EMQ, per-patient prior adaptation without temporal model; CC, no quantification.

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Table S6 MAE (pp) by reference burden stratum, pooled external cohorts (LightGBM)

Estimator	0 (AF-free) (n=813)	(0,1%] (n=16)	(1,10%] (n=30)	(10,50%] (n=35)	>50% (n=94)
CC	2.66	5.81	1.84	4.72	5.02
CC-MF	2.45	5.65	1.60	4.58	4.37
PCC-iso	3.47	6.89	3.31	5.66	6.08
ACC	2.59	5.88	2.06	5.10	4.45
EMQ	2.77	6.01	2.32	7.56	3.46
HMM-src	2.89	5.88	2.78	6.14	2.93
HMM-src[rawE]	2.22	5.62	1.53	4.54	4.16
PA-HMMQ (pre-registered)	2.91	5.95	2.77	6.18	2.90
PA-HMMQ (final)	2.24	5.62	1.56	4.44	4.00
PA-HMMQ[sel]	2.26	5.66	1.63	4.44	4.25

Mean over seeds.

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Table S7 Classification of patients by burden thresholds, pooled external cohorts (LightGBM)

Estimator	present_30s sens	present_30s spec	ge1pct sens	ge1pct spec	ge10pct sens	ge10pct spec
CC	0.992 ± 0.003	0.744 ± 0.034	0.992 ± 0.005	0.850 ± 0.026	0.974 ± 0.004	0.946 ± 0.011
CC-MF	0.982 ± 0.005	0.826 ± 0.027	0.989 ± 0.007	0.882 ± 0.022	0.974 ± 0.004	0.954 ± 0.010
PCC-iso	0.997 ± 0.003	0.487 ± 0.251	1.000 ± 0.000	0.712 ± 0.136	0.977 ± 0.000	0.923 ± 0.011
ACC	0.949 ± 0.028	0.846 ± 0.027	0.979 ± 0.008	0.879 ± 0.016	0.972 ± 0.004	0.948 ± 0.009
EMQ	0.961 ± 0.018	0.895 ± 0.025	0.976 ± 0.010	0.911 ± 0.020	0.975 ± 0.003	0.949 ± 0.006
HMM-src	0.971 ± 0.004	0.869 ± 0.018	0.987 ± 0.004	0.891 ± 0.019	0.978 ± 0.003	0.943 ± 0.009
HMM-src[rawE]	0.947 ± 0.014	0.910 ± 0.025	0.967 ± 0.011	0.925 ± 0.020	0.971 ± 0.003	0.961 ± 0.008
PA-HMMQ (pre-registered)	0.965 ± 0.008	0.877 ± 0.022	0.985 ± 0.006	0.889 ± 0.017	0.978 ± 0.003	0.940 ± 0.009
PA-HMMQ (final)	0.941 ± 0.014	0.914 ± 0.025	0.971 ± 0.012	0.923 ± 0.021	0.974 ± 0.004	0.960 ± 0.009
PA-HMMQ[sel]	0.941 ± 0.030	0.907 ± 0.046	0.967 ± 0.024	0.923 ± 0.038	0.974 ± 0.004	0.959 ± 0.011

present_30s: ≥30 s of AF in the analysed recording; ge1pct/ge10pct: burden ≥1%/≥10%. Mean ± SD over seeds.

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Table S8 Influence of atrial flutter (LightGBM, pooled external)

Estimator	MAE vs AF reference (pp)	MAE vs AF+AFL reference (pp)	MAE excluding patients with any AFL, n=949 (post hoc)
CC	2.99 ± 0.26	2.11 ± 0.22	1.67 ± 0.25
CC-MF	2.73 ± 0.23	1.85 ± 0.20	1.41 ± 0.23
PCC	3.94 ± 0.57	3.10 ± 0.67	2.67 ± 0.63
PCC-iso	3.85 ± 0.51	2.95 ± 0.57	2.55 ± 0.55
ACC	2.89 ± 0.22	1.98 ± 0.19	1.55 ± 0.21
PACC	3.21 ± 0.15	2.29 ± 0.17	1.87 ± 0.16
EMQ	3.04 ± 0.21	2.03 ± 0.20	1.66 ± 0.21
HMM-src	3.05 ± 0.29	2.02 ± 0.26	1.67 ± 0.28
HMM-src[rawE]	2.52 ± 0.22	1.65 ± 0.19	1.19 ± 0.22
PA-HMMQ (pre-registered)	3.07 ± 0.30	2.03 ± 0.27	1.69 ± 0.29
PA-HMMQ (final)	2.52 ± 0.24	1.64 ± 0.21	1.19 ± 0.23
PA-HMMQ[sel]	2.56 ± 0.27	1.69 ± 0.24	1.24 ± 0.27

The AF+AFL reference counts flutter time as arrhythmia time. The analysis excluding patients with any flutter is post hoc.

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Table S9 Robustness to simulated R-peak and rhythm perturbations, SHDB-AF (MAE in pp, mean ± SD over seeds)

Detector	Condition	CC	CC-MF	PCC-iso	ACC	EMQ	HMM-src	PA-HMMQ (pre-registered)	PA-HMMQ (final)
LightGBM	none_0.0	4.51 ± 0.12	4.31 ± 0.11	5.13 ± 0.42	4.73 ± 0.31	4.92 ± 0.52	4.53 ± 0.37	4.54 ± 0.36	4.23 ± 0.11
LightGBM	missed_0.02	4.60 ± 0.13	4.39 ± 0.11	5.49 ± 0.64	4.80 ± 0.31	5.33 ± 0.70	4.69 ± 0.35	4.71 ± 0.34	4.30 ± 0.09
LightGBM	missed_0.05	4.75 ± 0.13	4.55 ± 0.12	5.87 ± 0.72	4.96 ± 0.33	5.76 ± 0.85	4.94 ± 0.45	4.96 ± 0.44	4.47 ± 0.12
LightGBM	false_0.02	4.60 ± 0.16	4.33 ± 0.18	5.62 ± 0.59	4.80 ± 0.34	5.50 ± 0.81	4.82 ± 0.55	4.81 ± 0.58	4.18 ± 0.17
LightGBM	false_0.05	4.95 ± 0.29	4.53 ± 0.32	6.71 ± 0.77	5.10 ± 0.42	6.54 ± 1.15	5.44 ± 1.00	5.47 ± 1.10	4.27 ± 0.30
LightGBM	jitter_0.01	4.82 ± 0.15	4.53 ± 0.19	5.76 ± 0.55	5.03 ± 0.33	5.65 ± 0.72	5.07 ± 0.54	5.08 ± 0.55	4.40 ± 0.17
LightGBM	jitter_0.025	8.46 ± 1.12	7.53 ± 0.99	10.48 ± 1.06	8.62 ± 0.97	10.68 ± 0.88	9.61 ± 0.74	9.73 ± 0.81	6.79 ± 0.92
LightGBM	ectopy_0.05	4.68 ± 0.14	4.38 ± 0.12	5.63 ± 0.75	4.90 ± 0.31	5.49 ± 0.89	4.76 ± 0.57	4.78 ± 0.58	4.25 ± 0.13
LightGBM	ectopy_0.1	4.99 ± 0.22	4.56 ± 0.17	6.18 ± 0.84	5.17 ± 0.36	6.07 ± 1.17	5.13 ± 0.86	5.16 ± 0.88	4.34 ± 0.19
RRNet	none_0.0	4.78 ± 0.46	4.52 ± 0.40	5.00 ± 0.33	4.97 ± 0.44	4.85 ± 0.44	4.55 ± 0.41	4.60 ± 0.44	4.31 ± 0.28
RRNet	missed_0.02	6.13 ± 1.31	5.77 ± 1.24	6.58 ± 0.83	6.25 ± 1.25	6.74 ± 1.16	5.95 ± 0.92	6.03 ± 0.93	5.42 ± 1.07
RRNet	missed_0.05	7.49 ± 1.90	7.07 ± 1.91	8.17 ± 1.11	7.60 ± 1.85	8.82 ± 1.82	7.68 ± 1.39	7.72 ± 1.37	6.65 ± 1.74
RRNet	false_0.02	4.68 ± 0.44	4.40 ± 0.38	5.01 ± 0.29	4.86 ± 0.42	4.81 ± 0.46	4.47 ± 0.47	4.52 ± 0.47	4.17 ± 0.25
RRNet	false_0.05	4.62 ± 0.42	4.27 ± 0.35	5.16 ± 0.37	4.76 ± 0.39	4.85 ± 0.61	4.46 ± 0.66	4.53 ± 0.71	4.00 ± 0.25
RRNet	jitter_0.01	5.04 ± 0.52	4.76 ± 0.44	5.31 ± 0.39	5.20 ± 0.46	5.17 ± 0.51	4.82 ± 0.49	4.87 ± 0.51	4.55 ± 0.33
RRNet	jitter_0.025	7.91 ± 1.92	7.22 ± 1.62	8.65 ± 1.56	7.83 ± 1.72	8.24 ± 1.57	7.73 ± 1.92	7.81 ± 2.01	6.58 ± 1.38
RRNet	ectopy_0.05	5.49 ± 0.99	5.07 ± 0.91	5.81 ± 0.72	5.60 ± 0.98	5.75 ± 0.94	5.12 ± 0.81	5.19 ± 0.82	4.80 ± 0.71
RRNet	ectopy_0.1	6.47 ± 1.60	5.84 ± 1.45	6.87 ± 1.09	6.45 ± 1.59	6.93 ± 1.49	5.89 ± 1.13	6.00 ± 1.17	5.41 ± 1.24

missed/false: probability of deleting/inserting an R peak per beat; jitter: SD of Gaussian R-peak timing noise (s); ectopy: probability of a premature beat with compensatory pause.

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115 **Table S10 Coverage of normalised conformal intervals under perturbation, SHDB-AF**

Detector	Method	Condition	Coverage	Mean width (pp)
LightGBM	PA-HMMQ+conformal(norm)	none_0.0	0.845 ± 0.006	6.30 ± 0.70
LightGBM	PA-HMMQ[rawE]+conformal(norm)	none_0.0	0.843 ± 0.026	5.07 ± 1.22
LightGBM	PA-HMMQ[sel]+conformal(norm)	none_0.0	0.834 ± 0.037	4.85 ± 1.97
LightGBM	PA-HMMQ+conformal(norm)	missed_0.02	0.845 ± 0.006	6.55 ± 0.84
LightGBM	PA-HMMQ[rawE]+conformal(norm)	missed_0.02	0.847 ± 0.014	5.09 ± 1.26
LightGBM	PA-HMMQ[sel]+conformal(norm)	missed_0.02	0.837 ± 0.028	4.93 ± 2.07
LightGBM	PA-HMMQ+conformal(norm)	missed_0.05	0.854 ± 0.010	6.79 ± 0.88
LightGBM	PA-HMMQ[rawE]+conformal(norm)	missed_0.05	0.849 ± 0.008	5.32 ± 1.44
LightGBM	PA-HMMQ[sel]+conformal(norm)	missed_0.05	0.832 ± 0.035	5.19 ± 2.22
LightGBM	PA-HMMQ+conformal(norm)	false_0.02	0.856 ± 0.006	6.85 ± 0.95
LightGBM	PA-HMMQ[rawE]+conformal(norm)	false_0.02	0.849 ± 0.020	5.07 ± 1.19
LightGBM	PA-HMMQ[sel]+conformal(norm)	false_0.02	0.828 ± 0.041	5.18 ± 2.18
LightGBM	PA-HMMQ+conformal(norm)	false_0.05	0.834 ± 0.029	7.77 ± 1.29
LightGBM	PA-HMMQ[rawE]+conformal(norm)	false_0.05	0.865 ± 0.027	5.61 ± 1.41
LightGBM	PA-HMMQ[sel]+conformal(norm)	false_0.05	0.847 ± 0.046	5.43 ± 2.31
LightGBM	PA-HMMQ+conformal(norm)	jitter_0.01	0.849 ± 0.008	7.04 ± 0.86
LightGBM	PA-HMMQ[rawE]+conformal(norm)	jitter_0.01	0.839 ± 0.025	5.33 ± 1.20
LightGBM	PA-HMMQ[sel]+conformal(norm)	jitter_0.01	0.817 ± 0.054	5.21 ± 2.19
LightGBM	PA-HMMQ+conformal(norm)	jitter_0.025	0.682 ± 0.016	10.69 ± 1.14
LightGBM	PA-HMMQ[rawE]+conformal(norm)	jitter_0.025	0.746 ± 0.092	7.68 ± 1.48

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Detector	Method	Condition	Coverage	Mean width (pp)
LightGBM	PA-HMMQ[sel] +conformal(norm)	jitter_0.025	0.723 ± 0.124	7.29 ± 3.32
LightGBM	PA-HMMQ+conformal(norm)	ectopy_0.05	0.849 ± 0.008	6.67 ± 0.90
LightGBM	PA-HMMQ[rawE] +conformal(norm)	ectopy_0.05	0.852 ± 0.016	5.16 ± 1.20
LightGBM	PA-HMMQ[sel] +conformal(norm)	ectopy_0.05	0.830 ± 0.032	5.04 ± 2.08
LightGBM	PA-HMMQ+conformal(norm)	ectopy_0.1	0.841 ± 0.019	7.23 ± 1.03
LightGBM	PA-HMMQ[rawE] +conformal(norm)	ectopy_0.1	0.849 ± 0.027	5.37 ± 1.27
LightGBM	PA-HMMQ[sel] +conformal(norm)	ectopy_0.1	0.826 ± 0.044	5.31 ± 2.21
RRNet	PA-HMMQ+conformal(norm)	none_0.0	0.841 ± 0.009	6.83 ± 2.26
RRNet	PA-HMMQ[rawE] +conformal(norm)	none_0.0	0.841 ± 0.019	4.90 ± 1.38
RRNet	PA-HMMQ[sel] +conformal(norm)	none_0.0	0.826 ± 0.046	6.01 ± 4.36
RRNet	PA-HMMQ+conformal(norm)	missed_0.02	0.811 ± 0.050	8.01 ± 2.44
RRNet	PA-HMMQ[rawE] +conformal(norm)	missed_0.02	0.800 ± 0.055	5.58 ± 1.48
RRNet	PA-HMMQ[sel] +conformal(norm)	missed_0.02	0.785 ± 0.082	6.70 ± 4.61
RRNet	PA-HMMQ+conformal(norm)	missed_0.05	0.774 ± 0.058	9.07 ± 2.84
RRNet	PA-HMMQ[rawE] +conformal(norm)	missed_0.05	0.757 ± 0.077	6.19 ± 1.70
RRNet	PA-HMMQ[sel] +conformal(norm)	missed_0.05	0.744 ± 0.117	7.75 ± 5.82
RRNet	PA-HMMQ+conformal(norm)	false_0.02	0.847 ± 0.019	6.92 ± 2.24
RRNet	PA-HMMQ[rawE] +conformal(norm)	false_0.02	0.839 ± 0.020	4.90 ± 1.43
RRNet	PA-HMMQ[sel] +conformal(norm)	false_0.02	0.828 ± 0.049	6.07 ± 4.50
RRNet	PA-HMMQ+conformal(norm)	false_0.05	0.845 ± 0.028	7.10 ± 2.20
RRNet	PA-HMMQ[rawE] +conformal(norm)	false_0.05	0.834 ± 0.036	4.95 ± 1.33

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Detector	Method	Condition	Coverage	Mean width (pp)
RRNet	PA-HMMQ[sel] +conformal(norm)	false_0.05	0.822 ± 0.060	6.25 ± 4.68
RRNet	PA-HMMQ+conformal(norm)	jitter_0.01	0.843 ± 0.024	6.99 ± 2.24
RRNet	PA-HMMQ[rawE] +conformal(norm)	jitter_0.01	0.837 ± 0.028	5.00 ± 1.44
RRNet	PA-HMMQ[sel] +conformal(norm)	jitter_0.01	0.813 ± 0.058	6.24 ± 4.72
RRNet	PA-HMMQ+conformal(norm)	jitter_0.025	0.731 ± 0.064	9.76 ± 3.27
RRNet	PA-HMMQ[rawE] +conformal(norm)	jitter_0.025	0.731 ± 0.087	6.70 ± 1.85
RRNet	PA-HMMQ[sel] +conformal(norm)	jitter_0.025	0.735 ± 0.122	8.22 ± 5.90
RRNet	PA-HMMQ+conformal(norm)	ectopy_0.05	0.828 ± 0.040	7.57 ± 2.50
RRNet	PA-HMMQ[rawE] +conformal(norm)	ectopy_0.05	0.811 ± 0.048	5.32 ± 1.39
RRNet	PA-HMMQ[sel] +conformal(norm)	ectopy_0.05	0.796 ± 0.074	6.43 ± 4.59
RRNet	PA-HMMQ+conformal(norm)	ectopy_0.1	0.800 ± 0.054	8.25 ± 2.55
RRNet	PA-HMMQ[rawE] +conformal(norm)	ectopy_0.1	0.798 ± 0.066	5.75 ± 1.42
RRNet	PA-HMMQ[sel] +conformal(norm)	ectopy_0.1	0.787 ± 0.087	7.05 ± 4.96

Target 0.90.

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118 **Table S11 Sensitivity to training-set size and window length (LightGBM, MAE in pp)**

Setting	Estimator	Development test	SHDB-AF	CPSC2021
100% training data, W=60 (main)	CC	3.55 ± 1.05	4.51 ± 0.12	3.85 ± 0.76
100% training data, W=60 (main)	CC-MF	3.24 ± 1.22	4.31 ± 0.11	3.38 ± 0.67
100% training data, W=60 (main)	PCC-iso	4.16 ± 0.92	5.13 ± 0.42	5.50 ± 1.38
100% training data, W=60 (main)	ACC	3.49 ± 1.03	4.73 ± 0.31	3.53 ± 0.71
100% training data, W=60 (main)	EMQ	3.10 ± 0.93	4.92 ± 0.52	3.78 ± 1.12
100% training data, W=60 (main)	HMM-src	3.00 ± 1.02	4.53 ± 0.37	3.48 ± 1.02
100% training data, W=60 (main)	PA-HMMQ (pre-registered)	2.97 ± 1.03	4.54 ± 0.36	3.50 ± 1.06
100% training data, W=60 (main)	PA-HMMQ (final)	3.04 ± 1.26	4.23 ± 0.11	2.88 ± 0.39
25% training data	CC	3.45 ± 0.49	4.89 ± 0.16	4.02 ± 0.36
25% training data	CC-MF	2.99 ± 0.53	4.67 ± 0.16	3.46 ± 0.31
25% training data	PCC-iso	4.20 ± 0.82	5.36 ± 0.51	5.17 ± 0.91
25% training data	ACC	3.32 ± 0.45	5.12 ± 0.26	3.69 ± 0.41
25% training data	EMQ	2.88 ± 0.74	5.03 ± 0.38	3.23 ± 0.47
25% training data	HMM-src	2.48 ± 0.82	4.65 ± 0.10	3.17 ± 0.41
25% training data	PA-HMMQ (pre-registered)	2.56 ± 0.80	4.67 ± 0.09	3.24 ± 0.50
25% training data	PA-HMMQ (final)	2.71 ± 0.73	4.54 ± 0.15	2.92 ± 0.24
50% training data	CC	3.80 ± 1.40	4.89 ± 0.28	3.78 ± 0.43
50% training data	CC-MF	3.42 ± 1.53	4.66 ± 0.20	3.23 ± 0.37
50% training data	PCC-iso	4.44 ± 1.11	5.48 ± 0.53	5.52 ± 1.33
50% training data	ACC	3.66 ± 1.34	5.13 ± 0.38	3.37 ± 0.43
50% training data	EMQ	3.02 ± 0.84	5.18 ± 0.50	3.38 ± 0.45
50% training data	HMM-src	2.91 ± 0.96	4.71 ± 0.09	3.24 ± 0.39
50% training data	PA-HMMQ (pre-registered)	2.96 ± 1.03	4.73 ± 0.09	3.30 ± 0.48
50% training data	PA-HMMQ (final)	3.18 ± 1.62	4.56 ± 0.16	2.70 ± 0.25
window = 30 beats	CC	3.69 ± 1.49	5.13 ± 0.23	5.51 ± 0.80
window = 30 beats	CC-MF	3.19 ± 1.66	4.76 ± 0.24	4.39 ± 0.75
window = 30 beats	PCC-iso	5.15 ± 1.57	6.04 ± 0.65	7.80 ± 1.57
window = 30 beats	ACC	3.52 ± 1.49	5.33 ± 0.31	4.90 ± 0.80
window = 30 beats	EMQ	3.39 ± 0.92	5.55 ± 0.50	5.14 ± 1.26
window = 30 beats	HMM-src	3.07 ± 0.97	4.99 ± 0.37	4.68 ± 0.94
window = 30 beats	PA-HMMQ (pre-registered)	3.01 ± 0.99	5.01 ± 0.35	4.87 ± 0.91
window = 30 beats	PA-HMMQ (final)	2.86 ± 1.93	4.49 ± 0.19	3.25 ± 0.55

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Setting	Estimator	Development test	SHDB-AF	CPSC2021
window = 120 beats	CC	3.46 ± 0.92	4.44 ± 0.18	3.64 ± 0.85
window = 120 beats	CC-MF	3.36 ± 0.93	4.37 ± 0.19	3.38 ± 0.80
window = 120 beats	PCC-iso	4.26 ± 0.51	5.11 ± 0.59	5.00 ± 1.25
window = 120 beats	ACC	3.50 ± 0.98	4.99 ± 0.25	3.32 ± 1.00
window = 120 beats	EMQ	3.34 ± 0.93	4.78 ± 0.57	3.02 ± 1.23
window = 120 beats	HMM-src	3.41 ± 0.95	4.38 ± 0.20	3.12 ± 0.95
window = 120 beats	PA-HMMQ (pre-registered)	3.42 ± 0.95	4.39 ± 0.20	3.10 ± 1.02
window = 120 beats	PA-HMMQ (final)	3.25 ± 0.85	4.33 ± 0.19	2.89 ± 0.82

Reduced-data and window-length runs used SHDB-AF and CPSC2021 as external cohorts (Icentia11k omitted for computational reasons).

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Table S12 Subgroup MAE (pp) in SHDB-AF by sex and age (LightGBM)

Estimator	Subgroup	n	MAE (pp)
CC	female	42	3.78
CC	male	51	5.12
CC	age < 69	45	1.91
CC	age ≥ 69	48	6.96
PA-HMMQ (pre-registered)	female	42	3.30
PA-HMMQ (pre-registered)	male	51	5.57
PA-HMMQ (pre-registered)	age < 69	45	1.68
PA-HMMQ (pre-registered)	age ≥ 69	48	7.23
PA-HMMQ (final)	female	42	3.63
PA-HMMQ (final)	male	51	4.73
PA-HMMQ (final)	age < 69	45	1.73
PA-HMMQ (final)	age ≥ 69	48	6.58

Age split at the cohort median.

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Table S13 The 15 external patients with the largest absolute error of the final PA-HMMQ (LightGBM, seed-averaged)

patient	ae	err	true_burden	ectopy_sv	ectopy_v	afl	hours	db
icentia_p05828	100.000	100.000	0.000	0.000	0.000	1.000	5.274	icentia11k
icentia_p01292	100.000	100.000	0.000	0.000	0.017	1.000	6.351	icentia11k
icentia_p01304	100.000	100.000	0.000	0.000	0.031	1.000	1.236	icentia11k
icentia_p05149	100.000	100.000	0.000	0.000	0.023	1.000	8.921	icentia11k
icentia_p07973	99.991	99.991	0.000	0.000	0.000	1.000	5.662	icentia11k
icentia_p00921	97.229	97.229	0.000	0.000	0.162	1.000	2.271	icentia11k
icentia_p09800	91.539	91.539	0.000	0.000	0.000	1.000	10.668	icentia11k
icentia_p06999	87.350	87.350	0.000	0.417	0.006	0.000	7.884	icentia11k
cpsc_II_54	87.160	-87.160	1.000	0.000	0.001	0.000	0.708	cpsc2021
icentia_p08389	85.769	85.769	0.000	0.000	0.001	0.858	6.546	icentia11k
icentia_p10993	80.605	80.605	0.000	0.000	0.000	0.806	7.194	icentia11k
icentia_p03332	78.338	78.338	0.000	0.000	0.000	1.000	7.721	icentia11k
icentia_p06217	70.810	70.810	0.000	0.016	0.003	0.708	7.050	icentia11k
shdb_2148042	70.431	-70.431	1.000	0.000	0.000	0.000	23.816	shdb-af
shdb_5210865	69.129	69.129	0.000	0.000	0.000	0.723	23.996	shdb-af

ae, absolute error (pp); err, signed error (pp); true_burden, fraction; ectopy and AFL as fractions of beats/time.

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Table S14 Paired comparisons in the pooled external cohorts for all detectors and PA-HMMQ variants

Detector	Proposed	Cohort	Comparator	n patients	Mean AE proposed (pp)	Mean AE comparator (pp)	Mean paired diff (pp) [95% CI]	p (Holm)
Classical-LR	PA-HMMQ (final)	Pooled external	CC	988	7.44	8.54	-1.10 [-1.31, -0.92]	7.6e-36
Classical-LR	PA-HMMQ (final)	Pooled external	CC-MF	988	7.44	8.14	-0.71 [-0.85, -0.57]	8.8e-06
Classical-LR	PA-HMMQ (final)	Pooled external	PCC	988	7.44	13.90	-6.46 [-6.88, -6.03]	7.7e-119
Classical-LR	PA-HMMQ (final)	Pooled external	PCC-iso	988	7.44	11.70	-4.26 [-4.70, -3.85]	5e-103
Classical-LR	PA-HMMQ (final)	Pooled external	ACC	988	7.44	8.40	-0.96 [-1.15, -0.77]	0.0002
Classical-LR	PA-HMMQ (final)	Pooled external	PACC	988	7.44	10.37	-2.93 [-3.27, -2.59]	4.8e-54
Classical-LR	PA-HMMQ (final)	Pooled external	EMQ	988	7.44	9.95	-2.51 [-3.08, -1.99]	0.086
Classical-LR	PA-HMMQ (final)	Pooled external	HMM-src	988	7.44	10.23	-2.80 [-3.37, -2.23]	1.9e-86
Classical-LR	PA-HMMQ (final)	Pooled external	HMM-src[rawE] (post hoc, same emissions)	988	7.44	7.39	0.05 [0.00, 0.10]	1.2e-30
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	CC	988	10.47	8.54	1.93 [1.43, 2.44]	5.8e-13
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	CC-MF	988	10.47	8.14	2.33 [1.82, 2.85]	9.5e-55
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	PCC	988	10.47	13.90	-3.42 [-3.99, -2.82]	1.8e-50
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	PCC-iso	988	10.47	11.70	-1.23 [-1.67, -0.77]	7e-35
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	ACC	988	10.47	8.40	2.08 [1.59, 2.57]	1.3e-68
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	PACC	988	10.47	10.37	0.11 [-0.31, 0.51]	0.91
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	EMQ	988	10.47	9.95	0.52 [0.27, 0.75]	4.1e-50

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Detector	Proposed	Cohort	Comparator	n patients	Mean AE proposed (pp)	Mean AE comparator (pp)	Mean paired diff (pp) [95% CI]	p (Holm)
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	HMM-src	988	10.47	10.23	0.24 [0.18, 0.31]	0.04
Classical-LR	PA-HMMQ (pre-registered)	Pooled external	HMM-src (post hoc, same emissions)	988	10.47	10.23	0.24 [0.18, 0.31]	0.02
Classical-LR	PA-HMMQ[sel]	Pooled external	CC	988	9.79	8.54	1.26 [0.87, 1.66]	8.4e-09
Classical-LR	PA-HMMQ[sel]	Pooled external	CC-MF	988	9.79	8.14	1.65 [1.25, 2.06]	6.3e-51
Classical-LR	PA-HMMQ[sel]	Pooled external	PCC	988	9.79	13.90	-4.10 [-4.61, -3.57]	5e-67
Classical-LR	PA-HMMQ[sel]	Pooled external	PCC-iso	988	9.79	11.70	-1.90 [-2.27, -1.51]	6.8e-56
Classical-LR	PA-HMMQ[sel]	Pooled external	ACC	988	9.79	8.40	1.40 [1.02, 1.78]	2.2e-61
Classical-LR	PA-HMMQ[sel]	Pooled external	PACC	988	9.79	10.37	-0.57 [-0.91, -0.24]	0.0018
Classical-LR	PA-HMMQ[sel]	Pooled external	EMQ	988	9.79	9.95	-0.16 [-0.40, 0.06]	2.8e-25
Classical-LR	PA-HMMQ[sel]	Pooled external	HMM-src	988	9.79	10.23	-0.44 [-0.57, -0.31]	3.4e-25
Classical-LR	PA-HMMQ[sel]	Pooled external	HMM-src[sel] (post hoc, same emissions)	988	9.79	9.64	0.16 [0.11, 0.22]	0.054
LightGBM	PA-HMMQ (final)	Pooled external	CC	988	2.52	2.99	-0.47 [-0.56, -0.38]	1e-08
LightGBM	PA-HMMQ (final)	Pooled external	CC-MF	988	2.52	2.73	-0.21 [-0.27, -0.17]	0.23
LightGBM	PA-HMMQ (final)	Pooled external	PCC	988	2.52	3.94	-1.42 [-1.57, -1.28]	2.4e-128
LightGBM	PA-HMMQ (final)	Pooled external	PCC-iso	988	2.52	3.85	-1.33 [-1.50, -1.16]	2.2e-125
LightGBM	PA-HMMQ (final)	Pooled external	ACC	988	2.52	2.89	-0.37 [-0.46, -0.30]	0.23
LightGBM	PA-HMMQ (final)	Pooled external	PACC	988	2.52	3.21	-0.69 [-0.83, -0.57]	0.2
LightGBM	PA-HMMQ (final)	Pooled external	EMQ	988	2.52	3.04	-0.53 [-0.74, -0.33]	3.6e-21
LightGBM	PA-HMMQ (final)	Pooled external	HMM-src	988	2.52	3.05	-0.53 [-0.73, -0.35]	8.2e-94

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Detector	Proposed	Cohort	Comparator	n patients	Mean AE proposed (pp)	Mean AE comparator (pp)	Mean paired diff (pp) [95% CI]	p (Holm)
LightGBM	PA-HMMQ (final)	Pooled external	HMM-src[rawE] (post hoc, same emissions)	988	2.52	2.52	0.00 [-0.02, 0.02]	6.4e-56
LightGBM	PA-HMMQ (pre-registered)	Pooled external	CC	988	3.07	2.99	0.09 [-0.07, 0.26]	0.54
LightGBM	PA-HMMQ (pre-registered)	Pooled external	CC-MF	988	3.07	2.73	0.34 [0.17, 0.53]	1.1e-27
LightGBM	PA-HMMQ (pre-registered)	Pooled external	PCC	988	3.07	3.94	-0.87 [-1.04, -0.69]	6.2e-105
LightGBM	PA-HMMQ (pre-registered)	Pooled external	PCC-iso	988	3.07	3.85	-0.78 [-0.92, -0.62]	4e-107
LightGBM	PA-HMMQ (pre-registered)	Pooled external	ACC	988	3.07	2.89	0.18 [0.03, 0.34]	7.5e-20
LightGBM	PA-HMMQ (pre-registered)	Pooled external	PACC	988	3.07	3.21	-0.14 [-0.27, -0.00]	0.019
LightGBM	PA-HMMQ (pre-registered)	Pooled external	EMQ	988	3.07	3.04	0.03 [-0.07, 0.13]	3.4e-58
LightGBM	PA-HMMQ (pre-registered)	Pooled external	HMM-src	988	3.07	3.05	0.02 [0.00, 0.04]	2.9e-43
LightGBM	PA-HMMQ (pre-registered)	Pooled external	HMM-src (post hoc, same emissions)	988	3.07	3.05	0.02 [0.00, 0.04]	5.8e-44
LightGBM	PA-HMMQ[sel]	Pooled external	CC	988	2.56	2.99	-0.43 [-0.52, -0.34]	2.6e-07
LightGBM	PA-HMMQ[sel]	Pooled external	CC-MF	988	2.56	2.73	-0.17 [-0.22, -0.12]	0.0096
LightGBM	PA-HMMQ[sel]	Pooled external	PCC	988	2.56	3.94	-1.38 [-1.52, -1.24]	7.2e-127
LightGBM	PA-HMMQ[sel]	Pooled external	PCC-iso	988	2.56	3.85	-1.29 [-1.45, -1.12]	4e-123
LightGBM	PA-HMMQ[sel]	Pooled external	ACC	988	2.56	2.89	-0.33 [-0.41, -0.25]	0.0041
LightGBM	PA-HMMQ[sel]	Pooled external	PACC	988	2.56	3.21	-0.65 [-0.79, -0.53]	0.66
LightGBM	PA-HMMQ[sel]	Pooled external	EMQ	988	2.56	3.04	-0.48 [-0.71, -0.28]	3.4e-32

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Detector	Proposed	Cohort	Comparator	n patients	Mean AE proposed (pp)	Mean AE comparator (pp)	Mean paired diff (pp) [95% CI]	p (Holm)
LightGBM	PA-HMMQ[sel]	Pooled external	HMM-src	988	2.56	3.05	-0.49 [-0.70, -0.30]	5.8e-10
LightGBM	PA-HMMQ[sel]	Pooled external	HMM-src[sel] (post hoc, same emissions)	988	2.56	2.57	-0.01 [-0.04, 0.01]	7.8e-39
RRNet	PA-HMMQ (final)	Pooled external	CC	988	3.45	3.96	-0.51 [-0.59, -0.43]	1.5e-18
RRNet	PA-HMMQ (final)	Pooled external	CC-MF	988	3.45	3.72	-0.27 [-0.32, -0.23]	0.014
RRNet	PA-HMMQ (final)	Pooled external	PCC	988	3.45	4.37	-0.92 [-1.05, -0.81]	4.5e-120
RRNet	PA-HMMQ (final)	Pooled external	PCC-iso	988	3.45	4.22	-0.77 [-0.88, -0.67]	1.8e-114
RRNet	PA-HMMQ (final)	Pooled external	ACC	988	3.45	3.85	-0.40 [-0.48, -0.33]	0.82
RRNet	PA-HMMQ (final)	Pooled external	PACC	988	3.45	4.10	-0.65 [-0.76, -0.55]	0.014
RRNet	PA-HMMQ (final)	Pooled external	EMQ	988	3.45	3.63	-0.18 [-0.31, -0.08]	3.9e-29
RRNet	PA-HMMQ (final)	Pooled external	HMM-src	988	3.45	3.73	-0.28 [-0.38, -0.19]	4.1e-78
RRNet	PA-HMMQ (final)	Pooled external	HMM-src[rawE] (post hoc, same emissions)	988	3.45	3.43	0.02 [-0.00, 0.04]	9.2e-33
RRNet	PA-HMMQ (pre-registered)	Pooled external	CC	988	3.78	3.96	-0.18 [-0.28, -0.09]	5.3e-08
RRNet	PA-HMMQ (pre-registered)	Pooled external	CC-MF	988	3.78	3.72	0.05 [-0.03, 0.14]	8e-06
RRNet	PA-HMMQ (pre-registered)	Pooled external	PCC	988	3.78	4.37	-0.60 [-0.73, -0.48]	1.7e-103
RRNet	PA-HMMQ (pre-registered)	Pooled external	PCC-iso	988	3.78	4.22	-0.45 [-0.57, -0.33]	9.2e-91
RRNet	PA-HMMQ (pre-registered)	Pooled external	ACC	988	3.78	3.85	-0.08 [-0.16, 0.01]	1.2e-09
RRNet	PA-HMMQ (pre-registered)	Pooled external	PACC	988	3.78	4.10	-0.32 [-0.43, -0.22]	0.37

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Detector	Proposed	Cohort	Comparator	n patients	Mean AE proposed (pp)	Mean AE comparator (pp)	Mean paired diff (pp) [95% CI]	p (Holm)
RRNet	PA-HMMQ (pre-registered)	Pooled external	EMQ	988	3.78	3.63	0.14 [0.06, 0.22]	1.3e-64
RRNet	PA-HMMQ (pre-registered)	Pooled external	HMM-src	988	3.78	3.73	0.04 [0.02, 0.07]	3.4e-21
RRNet	PA-HMMQ (pre-registered)	Pooled external	HMM-src (post hoc, same emissions)	988	3.78	3.73	0.04 [0.02, 0.07]	6.8e-22
RRNet	PA-HMMQ[sel]	Pooled external	CC	988	3.50	3.96	-0.46 [-0.54, -0.38]	1.2e-15
RRNet	PA-HMMQ[sel]	Pooled external	CC-MF	988	3.50	3.72	-0.23 [-0.28, -0.17]	0.63
RRNet	PA-HMMQ[sel]	Pooled external	PCC	988	3.50	4.37	-0.88 [-1.00, -0.76]	7.6e-117
RRNet	PA-HMMQ[sel]	Pooled external	PCC-iso	988	3.50	4.22	-0.73 [-0.83, -0.63]	7.7e-111
RRNet	PA-HMMQ[sel]	Pooled external	ACC	988	3.50	3.85	-0.35 [-0.43, -0.28]	0.63
RRNet	PA-HMMQ[sel]	Pooled external	PACC	988	3.50	4.10	-0.60 [-0.70, -0.50]	0.16
RRNet	PA-HMMQ[sel]	Pooled external	EMQ	988	3.50	3.63	-0.14 [-0.27, -0.02]	3.4e-36
RRNet	PA-HMMQ[sel]	Pooled external	HMM-src	988	3.50	3.73	-0.23 [-0.34, -0.14]	2.6e-06
RRNet	PA-HMMQ[sel]	Pooled external	HMM-src[sel] (post hoc, same emissions)	988	3.50	3.49	0.01 [-0.02, 0.04]	4.3e-25

As Table 4. Comparisons for each individual cohort and the development test set are provided in the code repository (tables/tableS_all_tests.csv).

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Table S15 Revision R1 (post hoc): emission model selected on calibration-A patients per seed and detector

Seed	Detector	Selected emission	calA MAE hist (pp)	calA MAE T=0.5 (pp)	calA MAE T=1 (pp)	calA MAE T=2 (pp)	calA MAE T=4 (pp)	calA MAE T=8 (pp)
11	Classical-LR	hist	7.92	9.53	9.60	9.73	9.49	9.31
11	LightGBM	T=0.5	7.49	6.50	6.71	7.26	7.69	7.62
11	RRNet	T=0.5	6.34	5.78	6.66	6.96	7.07	7.05
22	Classical-LR	T=4	7.37	7.50	7.29	7.34	6.96	7.06
22	LightGBM	T=8	5.77	6.14	6.11	5.91	5.75	5.72
22	RRNet	T=4	3.15	3.03	3.15	3.12	2.95	3.09
33	Classical-LR	hist	7.55	8.01	8.14	8.21	7.94	7.93
33	LightGBM	T=0.5	0.88	0.65	0.76	0.92	1.09	1.12
33	RRNet	T=1	0.66	0.54	0.47	0.69	0.89	0.98
44	Classical-LR	hist	3.42	6.50	6.32	6.03	5.55	5.26
44	LightGBM	T=0.5	0.86	0.48	0.53	0.68	1.14	1.03
44	RRNet	T=0.5	1.26	1.09	1.26	1.31	1.36	1.36
55	Classical-LR	hist	7.34	9.42	9.52	9.83	9.77	9.60
55	LightGBM	T=4	2.45	1.45	1.39	1.20	1.01	1.05
55	RRNet	T=8	2.34	1.98	2.14	2.19	1.63	1.20

Mean per-patient burden error of the fixed-dynamics HMM on calibration-A patients for each candidate emission (hist: calibration-set histogram; T: likelihood ratio (p/(1−p))^(1/T)).

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Table S16 External patients with ≥30 s of reference AF classified as AF-present by classify-and-count but AF-free by the final PA-HMMQ (LightGBM, seed-majority; post hoc)

patient	db	true_burden	af_minutes	hours
shdb_4878813	shdb-af	0.0061	8.7389	23.9871
shdb_5113281	shdb-af	0.0008	1.1737	23.9804
shdb_5218251	shdb-af	0.002	2.8342	23.8223
shdb_5494269	shdb-af	0.0008	1.109	22.8142
shdb_5577891	shdb-af	0.0006	0.8888	23.9025
cpsc_II_64	cpsc2021	0.0233	1.5813	1.1315
cpsc_II_66	cpsc2021	0.0361	4.7462	2.1907
cpsc_II_92	cpsc2021	0.0346	4.1198	1.9867
cpsc__I_31	cpsc2021	0.0072	2.4503	5.7086

true_burden as a fraction of analysed time; af_minutes, reference AF time; hours, analysed time.

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Table S17 Post-hoc paired comparisons with burden estimates reported at 0.1-percentage-point resolution (LightGBM, pooled external cohorts)

Proposed	Comparator	Mean AE proposed (pp)	Mean paired diff (pp) [95% CI]	Non-zero pairs	Rank-biserial r	p (Holm)
PA-HMMQ (final)	CC	2.52	-0.47 [-0.56, -0.38]	444	-0.733	4.1e-40
PA-HMMQ (final)	CC-MF	2.52	-0.22 [-0.27, -0.17]	326	-0.699	2.3e-27
PA-HMMQ (final)	PCC	2.52	-1.42 [-1.57, -1.28]	988	-0.888	1.7e-129
PA-HMMQ (final)	PCC-iso	2.52	-1.33 [-1.50, -1.16]	984	-0.878	6.8e-125
PA-HMMQ (final)	ACC	2.52	-0.37 [-0.46, -0.30]	321	-0.737	9.6e-30
PA-HMMQ (final)	PACC	2.52	-0.70 [-0.83, -0.57]	355	-0.790	2.3e-37
PA-HMMQ (final)	EMQ	2.52	-0.53 [-0.74, -0.33]	252	-0.495	1e-11
PA-HMMQ (final)	HMM-src	2.52	-0.53 [-0.73, -0.35]	266	-0.660	2.2e-20
PA-HMMQ (final)	HMM-src[rawE]	2.52	0.00 [-0.02, 0.02]	192	0.111	0.18
PA-HMMQ[sel]	CC	2.56	-0.43 [-0.52, -0.34]	473	-0.670	9.4e-36
PA-HMMQ[sel]	CC-MF	2.56	-0.17 [-0.23, -0.13]	371	-0.544	3.1e-19
PA-HMMQ[sel]	PCC	2.56	-1.38 [-1.52, -1.24]	988	-0.883	5.9e-128
PA-HMMQ[sel]	PCC-iso	2.56	-1.29 [-1.45, -1.12]	985	-0.869	1.1e-122
PA-HMMQ[sel]	ACC	2.56	-0.33 [-0.42, -0.26]	367	-0.594	2.6e-22
PA-HMMQ[sel]	PACC	2.56	-0.65 [-0.79, -0.53]	391	-0.706	5.7e-33
PA-HMMQ[sel]	EMQ	2.56	-0.48 [-0.71, -0.28]	318	-0.212	0.0011
PA-HMMQ[sel]	HMM-src	2.56	-0.49 [-0.70, -0.30]	320	-0.353	9.1e-08
PA-HMMQ[sel]	HMM-src[sel]	2.56	-0.01 [-0.04, 0.01]	337	0.003	0.96

Same procedure as Table 4 after rounding each estimate to 0.001 (0.1 pp); fair comparator (same emissions, fixed dynamics) not included in the Holm family.

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Table S18 Post-hoc local recalibration of the normalised conformal quantile in each external cohort (final PA-HMMQ, LightGBM)

Cohort	Local calibration patients	Coverage, mean	Coverage, 5th–95th percentile	Coverage in AF patients	Mean width (pp)
CPSC2021	10	0.910	0.747–1.000	0.886	10.54
CPSC2021	20	0.903	0.765–0.988	0.876	7.45
CPSC2021	40	0.901	0.785–0.985	0.873	6.73
Icentia11k	10	0.913	0.758–0.996	0.816	14.75
Icentia11k	20	0.908	0.790–0.983	0.814	3.65
Icentia11k	40	0.903	0.817–0.967	0.803	0.91
SHDB-AF	10	0.907	0.723–1.000	0.881	22.28
SHDB-AF	20	0.907	0.781–0.986	0.881	17.02
SHDB-AF	40	0.904	0.792–0.981	0.876	13.97

For each seed, n locally labelled patients were drawn at random (200 repeats) to compute the conformal quantile; coverage and width evaluated on the remaining patients of the cohort. Target coverage 0.90.

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Table S19 Post-hoc comparison of median-filter lengths for classify-and-count (LightGBM, mean over seeds)

Median-filter length (windows)	Development MAE (pp)	Pooled external MAE (pp)
3	3.00	2.73
5	2.96	2.66
7	2.91	2.63
9	2.94	2.60
15	3.00	2.56

Filter length selected on development patients (calibration B and development test) before inspecting the external results of this analysis.

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Figure legends

Figure S1 Distribution of reference AF burden (A) and analysed monitoring time per patient (B) in the development test set (seed 11) and the external cohorts.

Alt text: Two cumulative distribution plots, one per panel, with four lines for the cohorts.

Figure S2 Window-level ROC (top; x-axis truncated at 0.3) and precision–recall (bottom) curves of the three detectors in each cohort (seed 11).

Alt text: Grid of eight small line charts showing ROC and precision-recall curves for three detectors in four cohorts.

Figure S3 Row-normalised window-level confusion matrices of the LightGBM detector (all seeds pooled).

Alt text: Four two-by-two heat maps with proportions of correctly and incorrectly classified AF and non-AF windows.

Figure S4 RRNet training and inner-validation binary cross-entropy per epoch for the five seeds.

Alt text: Line chart of loss against epoch, ten lines for five seeds.

Figure S5 Mean absolute burden error as a function of the prior strength κ for calibrated-histogram and probability emissions, in the development test set and pooled external cohorts (LightGBM; mean ± SD over seeds).

Alt text: Line chart with six κ values on the horizontal axis and error on the vertical axis for four combinations of emission model and cohort.

Figure S6 Mean paired difference in per-patient absolute burden error between the final PA-HMMQ and each comparator in the pooled external cohorts (LightGBM), with bootstrap 95% CI and Holm-adjusted P values.

Alt text: Forest plot with eight comparators, all point estimates left of zero.

Figure S7 Per-patient absolute error of the final PA-HMMQ versus the proportion of ectopic beats (A) and atrial flutter burden (B), pooled external cohorts (log–log axes).

Alt text: Two scatter plots on logarithmic axes; large errors cluster at high flutter burden.

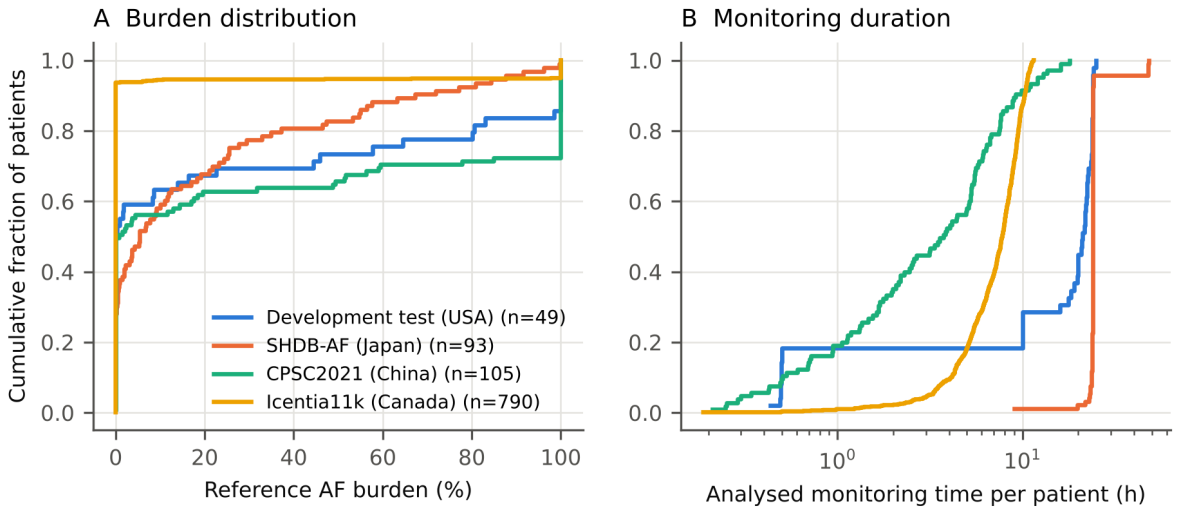
Figure S8 Median CPU time per 24-h recording of each pipeline component.

31

Alt text: Horizontal bar chart of milliseconds per 24-hour recording for feature extraction, detector inference and burden estimators on a logarithmic axis.

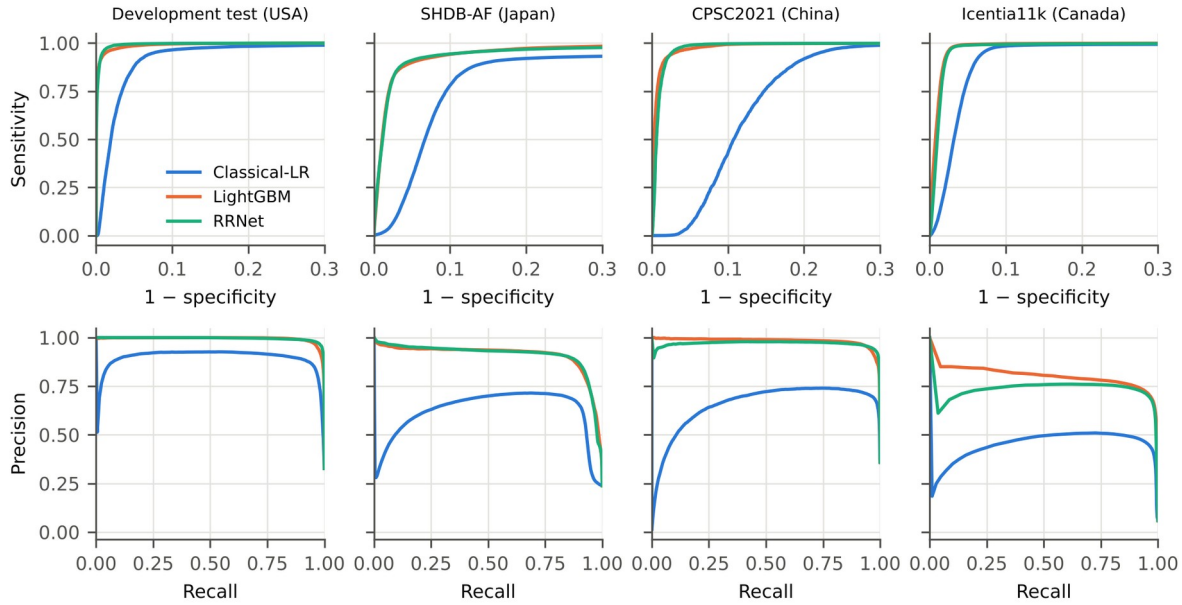
32

Figure S1


A Burden distribution
B Monitoring duration
Cumulative fraction of patients
Reference AF burden (%)
Analysed monitoring time per patient (h)
Development test (USA) (n=49)
SHDB-AF (Japan) (n=93)
CPSC2021 (China) (n=105)
Icentia11k (Canada) (n=790)

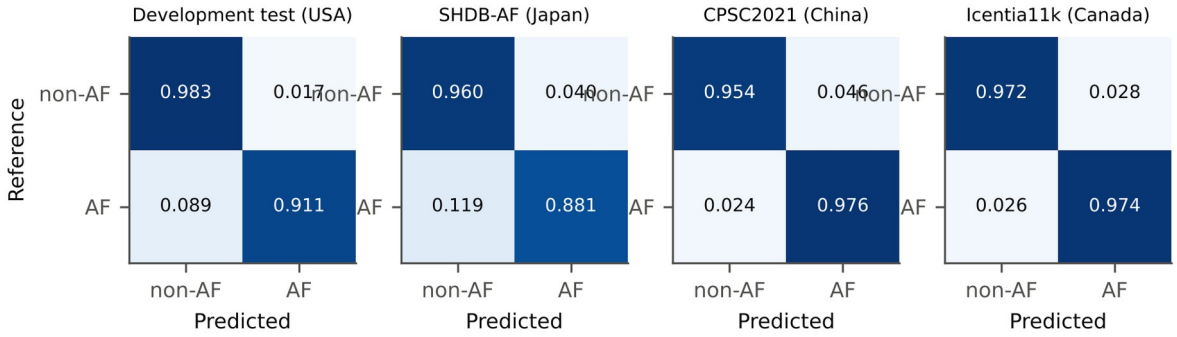
33

Figure S2


Development test (USA)
SHDB-AF (Japan)
CPSC2021 (China)
Icentia11k (Canada)
Sensitivity
Classical-LR
LightGBM
RRNet
1 − specificity
Precision
Recall
0.00
0.25
0.50
0.75
1.00
0.0
0.1
0.2
0.3

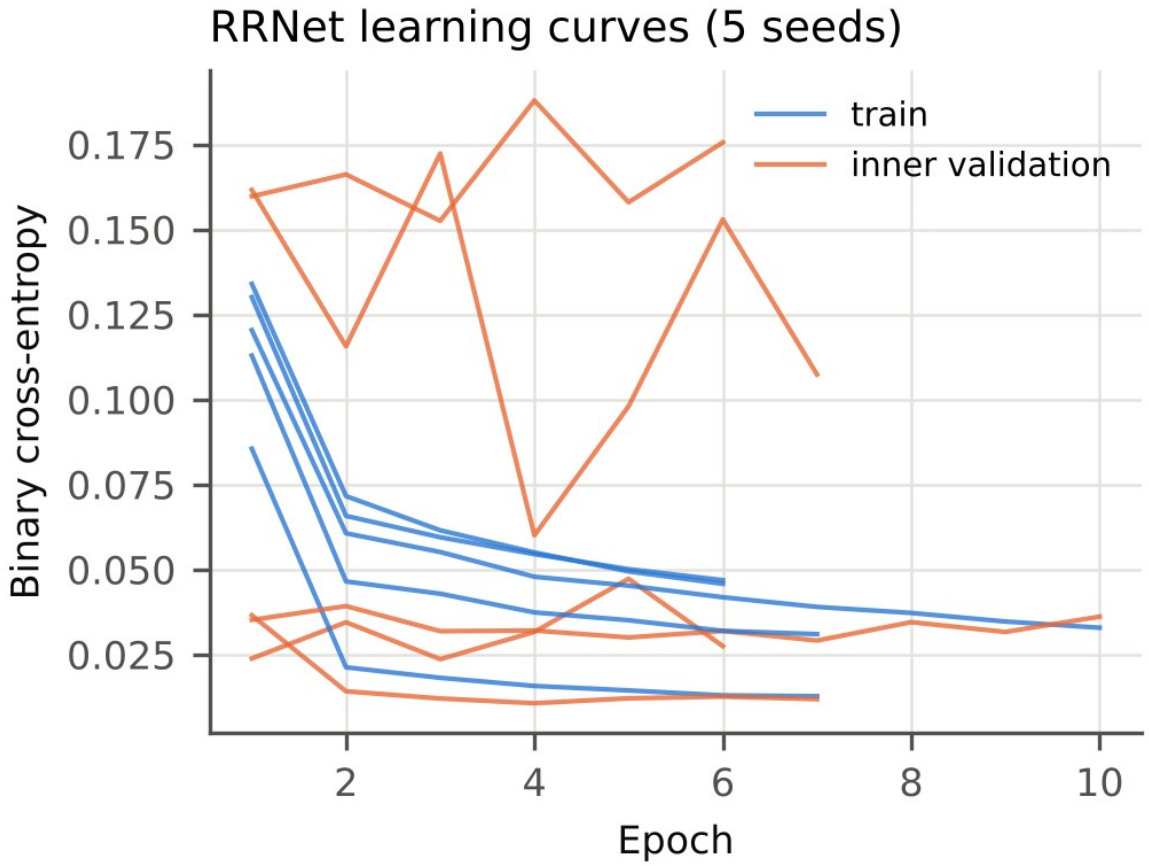
34

Figure S3


Development test (USA)
SHDB-AF (Japan)
CPSC2021 (China)
Icentia11k (Canada)
Reference
non-AF
AF
0.983
0.017
0.089
0.911
0.960
0.040
0.119
0.881
0.954
0.046
0.024
0.976
0.972
0.028
0.026
0.974
non-AF
AF
Predicted

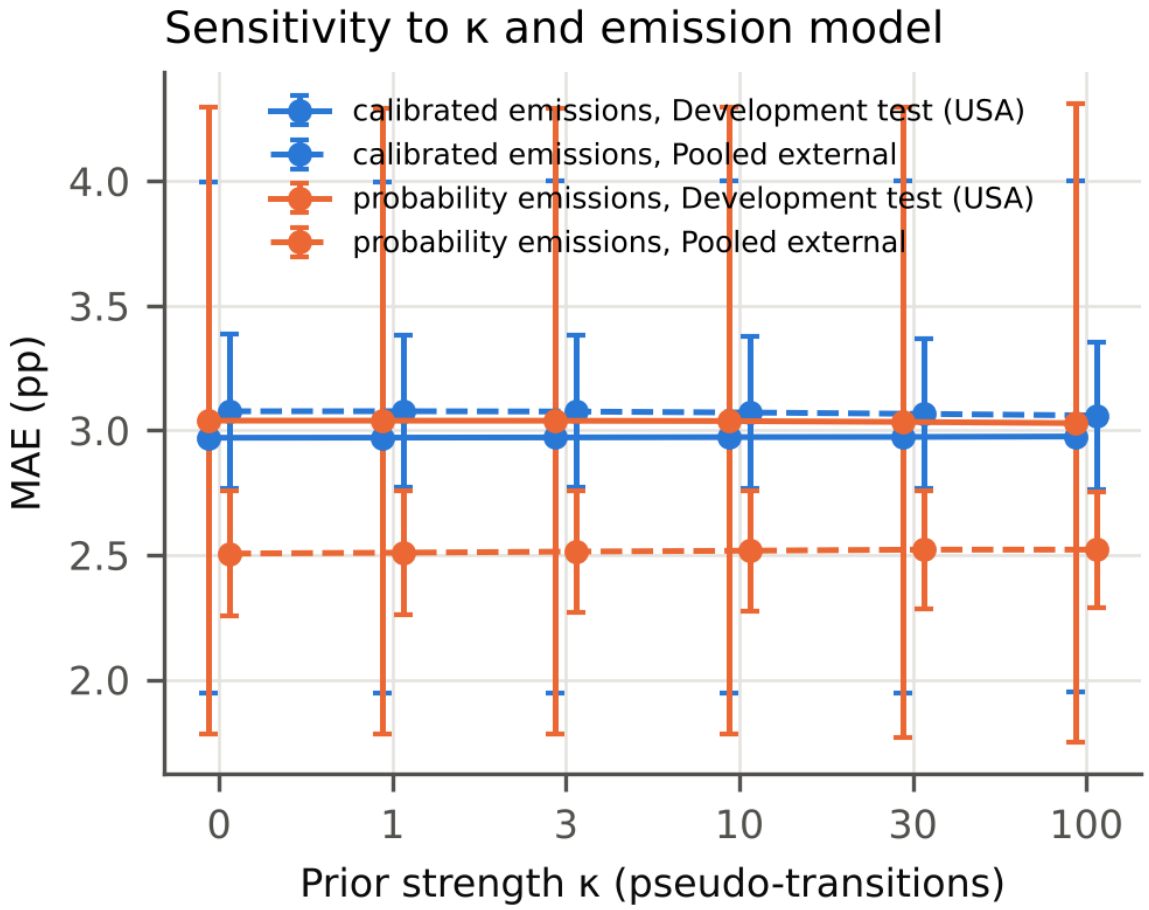
35

Figure S4


RRNet learning curves (5 seeds)
train
inner validation
Binary cross-entropy
0.175
0.150
0.125
0.100
0.075
0.050
0.025
2
4
6
8
10
Epoch

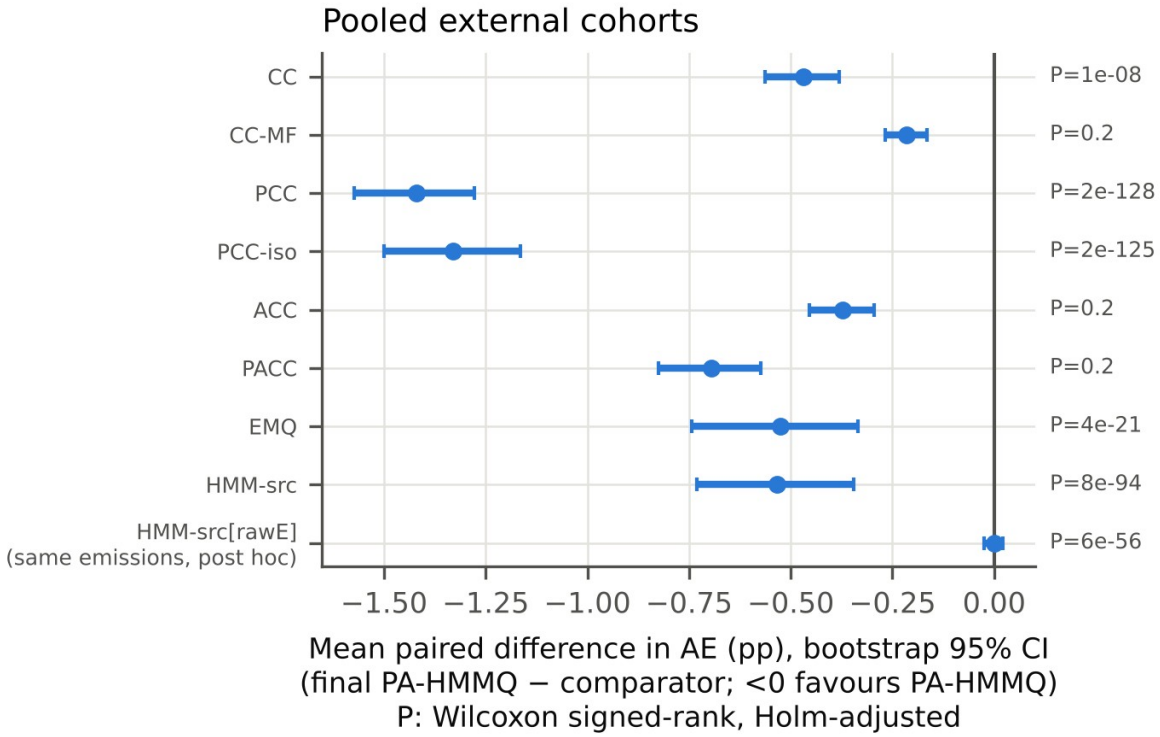
36

Figure S5


Sensitivity to κ and emission model
calibrated emissions, Development test (USA)
calibrated emissions, Pooled external
probability emissions, Development test (USA)
probability emissions, Pooled external
MAE (pp)
4.0
3.5
3.0
2.5
2.0
0
1
3
10
30
100
Prior strength κ (pseudo-transitions)

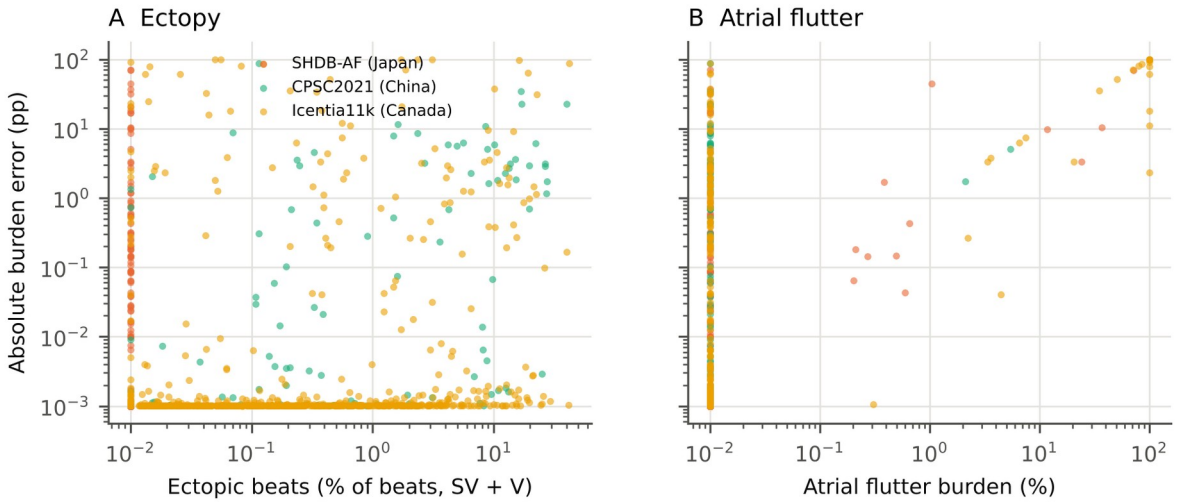
37

Figure S6


Pooled external cohorts
CC
CC-MF
PCC
PCC-iso
ACC
PACC
EMQ
HMM-src
HMM-src[rawE]
(same emissions, post hoc)
P=1e-08
P=0.2
P=2e-128
P=2e-125
P=0.2
P=0.2
P=4e-21
P=8e-94
P=6e-56
−1.50 −1.25 −1.00 −0.75 −0.50 −0.25 0.00
Mean paired difference in AE (pp), bootstrap 95% CI
(final PA-HMMQ − comparator; <0 favours PA-HMMQ)
P: Wilcoxon signed-rank, Holm-adjusted

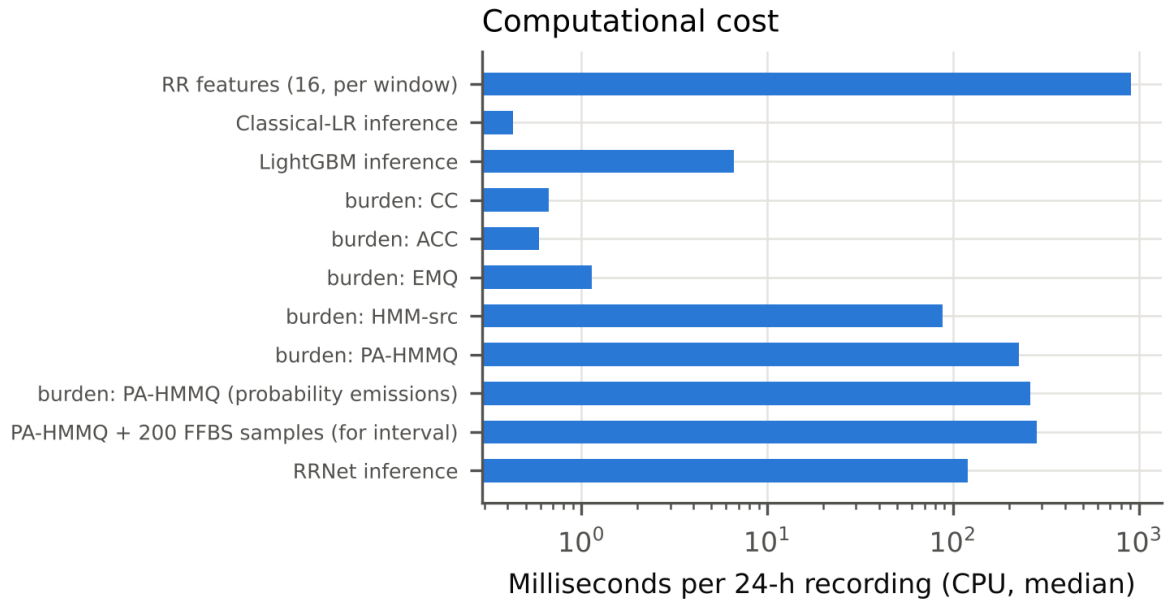
38

Figure S7


A Ectopy
B Atrial flutter
SHDB-AF (Japan)
CPSC2021 (China)
Icentia11k (Canada)
Absolute burden error (pp)
Ectopic beats (% of beats, SV + V)
Atrial flutter burden (%)

39

Figure S8


Computational cost
RR features (16, per window)
Classical-LR inference
LightGBM inference
burden: CC
burden: ACC
burden: EMQ
burden: HMM-src
burden: PA-HMMQ
burden: PA-HMMQ (probability emissions)
PA-HMMQ + 200 FFBS samples (for interval)
RRNet inference
10^0
10^1
10^2
10^3
Milliseconds per 24-h recording (CPU, median)

40